

Wachirawit Piyaprapapan

AI Engineer | ML Systems | RAG | Edge AI | Data Platforms

Bangkok, Thailand | wachirawit.p.work@gmail.com | 098 828 2779 | linkedin.com/in/wachirawit-piyaprapapan | github.com/KheawKachee

Education

Chulalongkorn University, B.Eng. in Electrical Engineering – Bangkok, Thailand Aug 2022 – May 2026

- GPAX: 3.44 / 4.00; Second-class honours.
- Relevant coursework: Data Science, Data Engineering, Estimation, Statistical Learning, Optimization.
- Capstone: Generative Video-Based Sky Image Forecasting for Thai sky images using VQ-VAE-based video prediction.

Skills

Languages: Python, SQL, JavaScript, Bash**AI / ML Systems:** PyTorch, OpenCV, Hugging Face, spaCy, RAG, LiteLLM, Pydantic, time-series forecasting, computer vision**Backend / Data Systems:** FastAPI, Next.js, React, PostgreSQL, SQLAlchemy, Pandas, PySpark, Airflow, dbt**MLOps / Deployment:** Docker, Git, Grafana, Prometheus, GCP, Vercel, Selenium

Experience

AI Engineer, Part-time, Hobbit Technologies – Bangkok, Thailand Feb 2026 – present

- Develop and maintain production-oriented RAG services with LiteLLM, FastAPI, and Pydantic-validated structured outputs to reduce downstream parsing failures.
- Architect a modular Next.js/PostgreSQL platform for CEFR-aligned content generation, automated grammar correction, and evaluation workflows.
- Improve LLM reliability through prompt templates, schema validation, REST API service boundaries, and reusable backend components.

AI Engineer Intern, Hobbit Technologies – Bangkok, Thailand June 2025 – Aug 2025

- Engineered a Dockerized computer-vision data engine with OpenCV, automated drift checks, and active-learning triggers, reducing labeling cost by approximately 20k THB/year.
- Deployed Grafana/Prometheus monitoring for Edge AI inference on on-premise IoT sensors, improving visibility into production-line failures.
- Built data-cleaning and dataset-versioning workflows for edge computer-vision experiments.

Projects

Generative Video-Based Sky Image Forecasting for Thai Sky Images

- Implemented and optimized a VQ-VAE-based forecasting model with Charbonnier loss, perceptual loss, EMA codebook updates, and sub-pixel interpolation decoding.
- Achieved 37.32 dB peak validation PSNR, a 16% improvement over baseline, while compressing channel-sequence representations for more stable temporal prediction.

Coffee Chain Demand Forecasting System

- Engineered horizon-safe lag, rolling, event-proximity, calendar, and store-category demand features for daily product-level sales forecasting across representative coffee stores.
- Reduced internal walk-forward CV MAE by 12.2% over the leakage-safe baseline and identified local-event and store-category behavior as the strongest business signals.

Competitions

Microsoft AI Engineering Skills and Hackathon for Employment 2026 2026

- Reached finalist team stage among 2,000+ participants.
- Built a dashboard-driven emergency-case intake and evaluation pipeline using Azure Foundry and Azure OpenAI.

Thailand Super AI Engineer Season 6 — Edge AI for Intelligent Transport Systems 2026

- Developed a computer-vision data pipeline for traffic monitoring and anomaly detection.
- Used VLM-assisted inspection and semi-supervised learning techniques to bootstrap external public datasets.

Thailand Super AI Engineer Season 6 — Coffee Chain Time-Series Hackathon 2026

- Led feature engineering for daily coffee-sales forecasting and connected model outputs to inventory and order-optimization workflows.
- Designed leakage-safe validation and event-driven demand features for production-like forecasting constraints.

I-squared Hackathon — Motorbike Rider Anomaly Detection & Classification

- Built a two-stage computer-vision pipeline using YOLOv8 for detection and ViT for classification.
- Designed automated data-cleaning and image-enhancement workflows; reached semifinals with over 90% training accuracy and over 70% accuracy on unseen real-world images.